

Exploration in RL: Multi-armed Bandit

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CS 1840 Fall 2026

Announcements

HW1 is out!

Check the course website for Office Hours:
<https://harvard-cs-1840.github.io/2026/>

Recap: Value Iteration vs Policy Iteration

Policy Iteration:

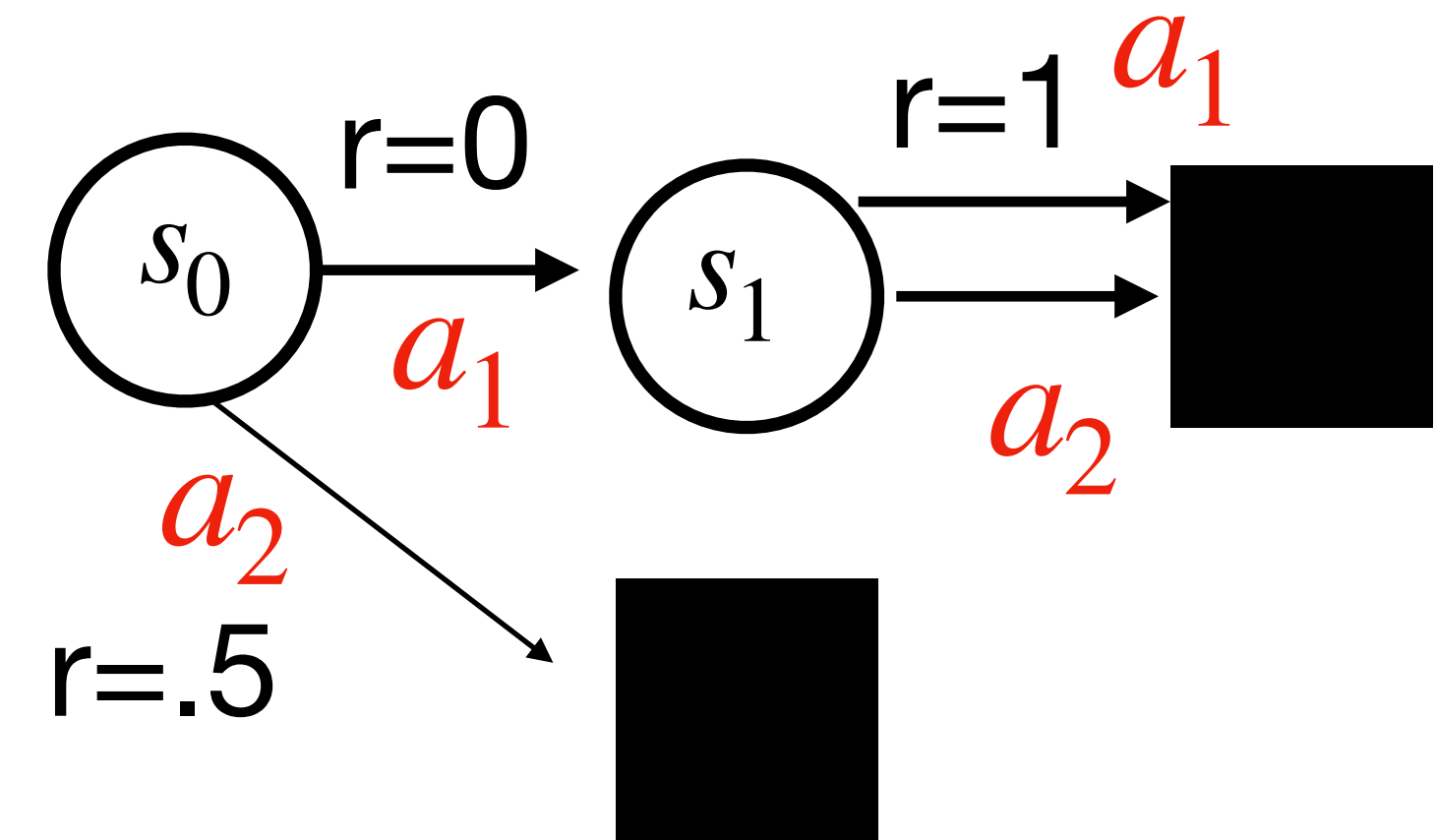
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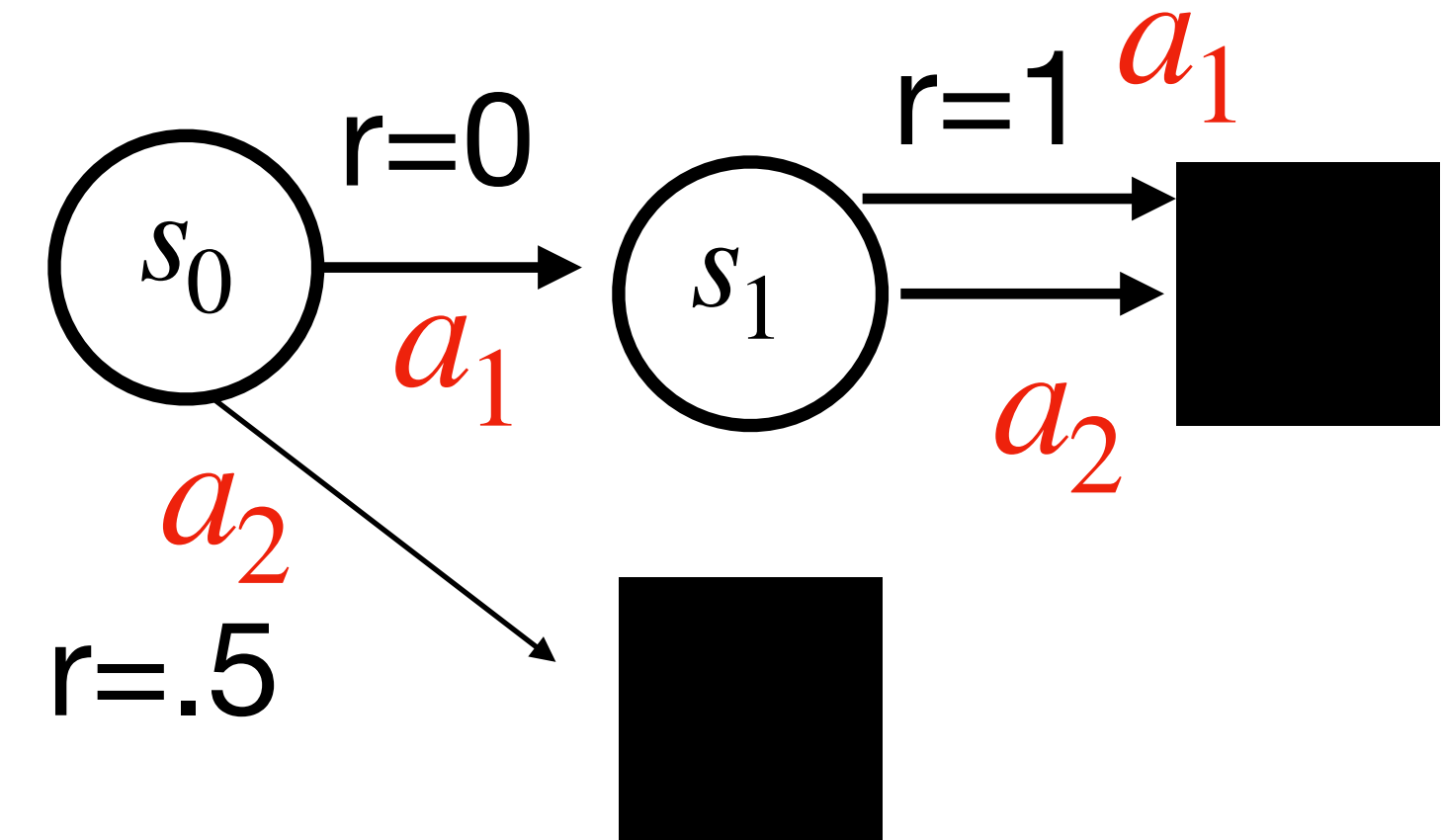
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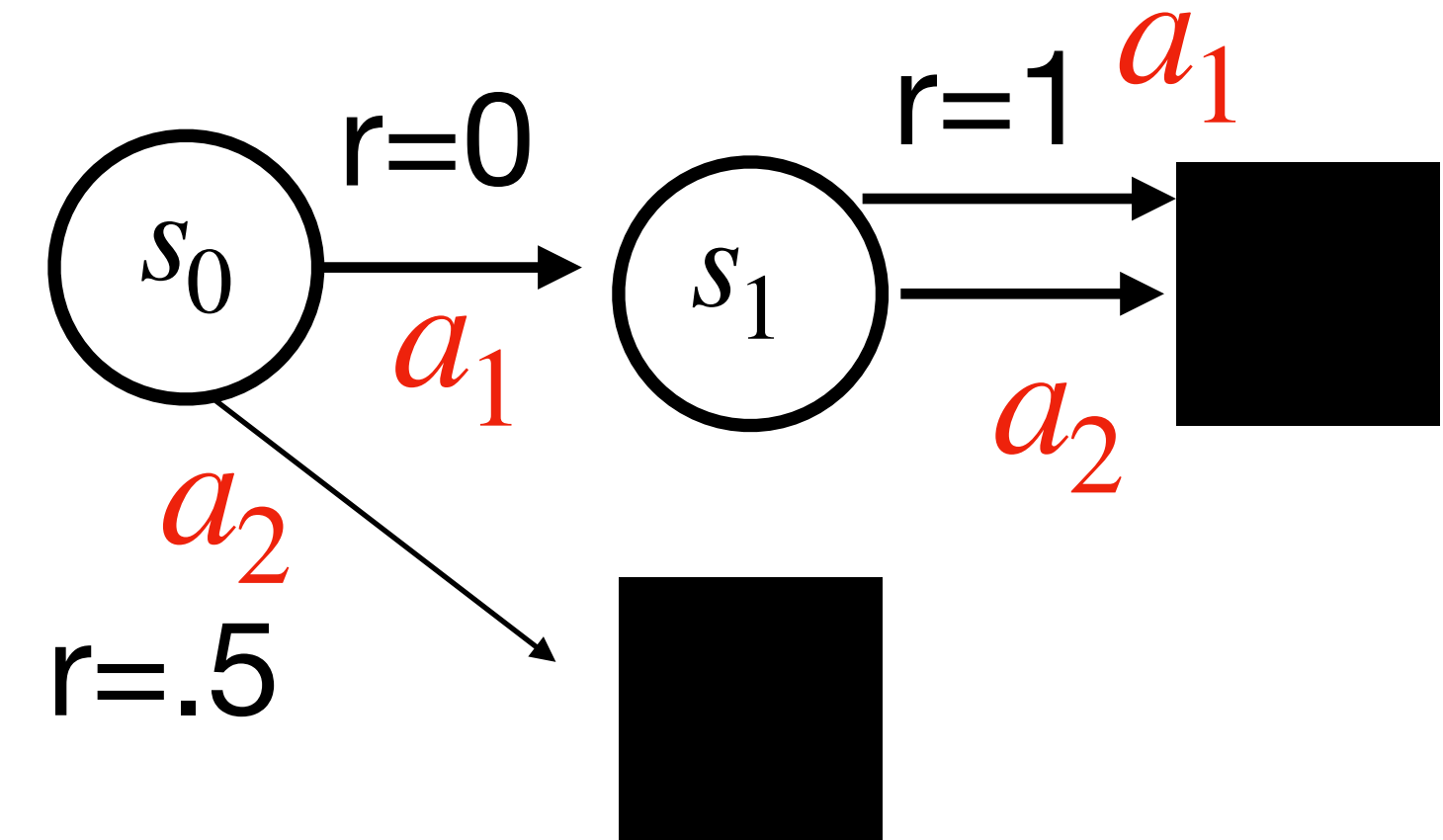
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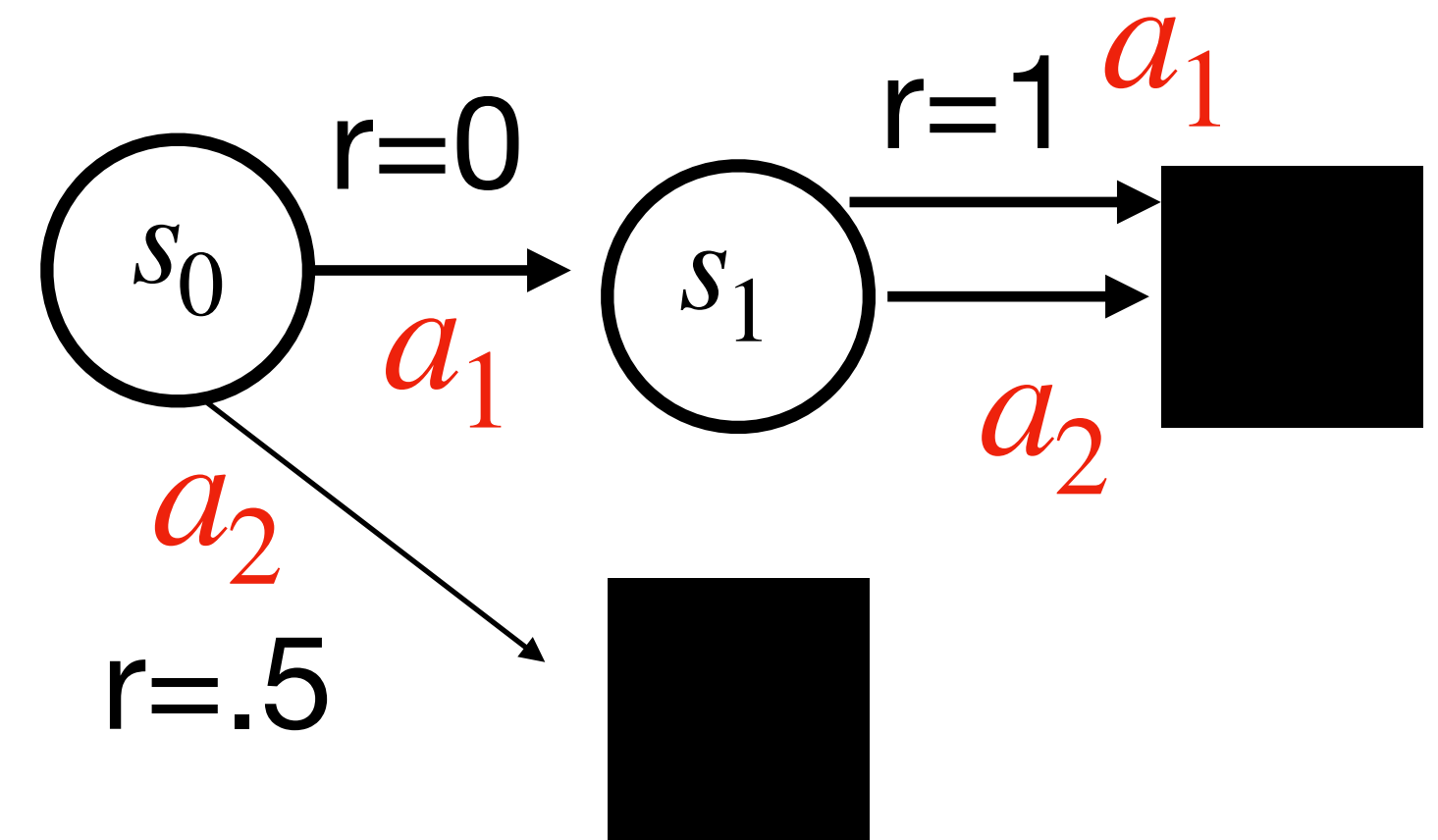
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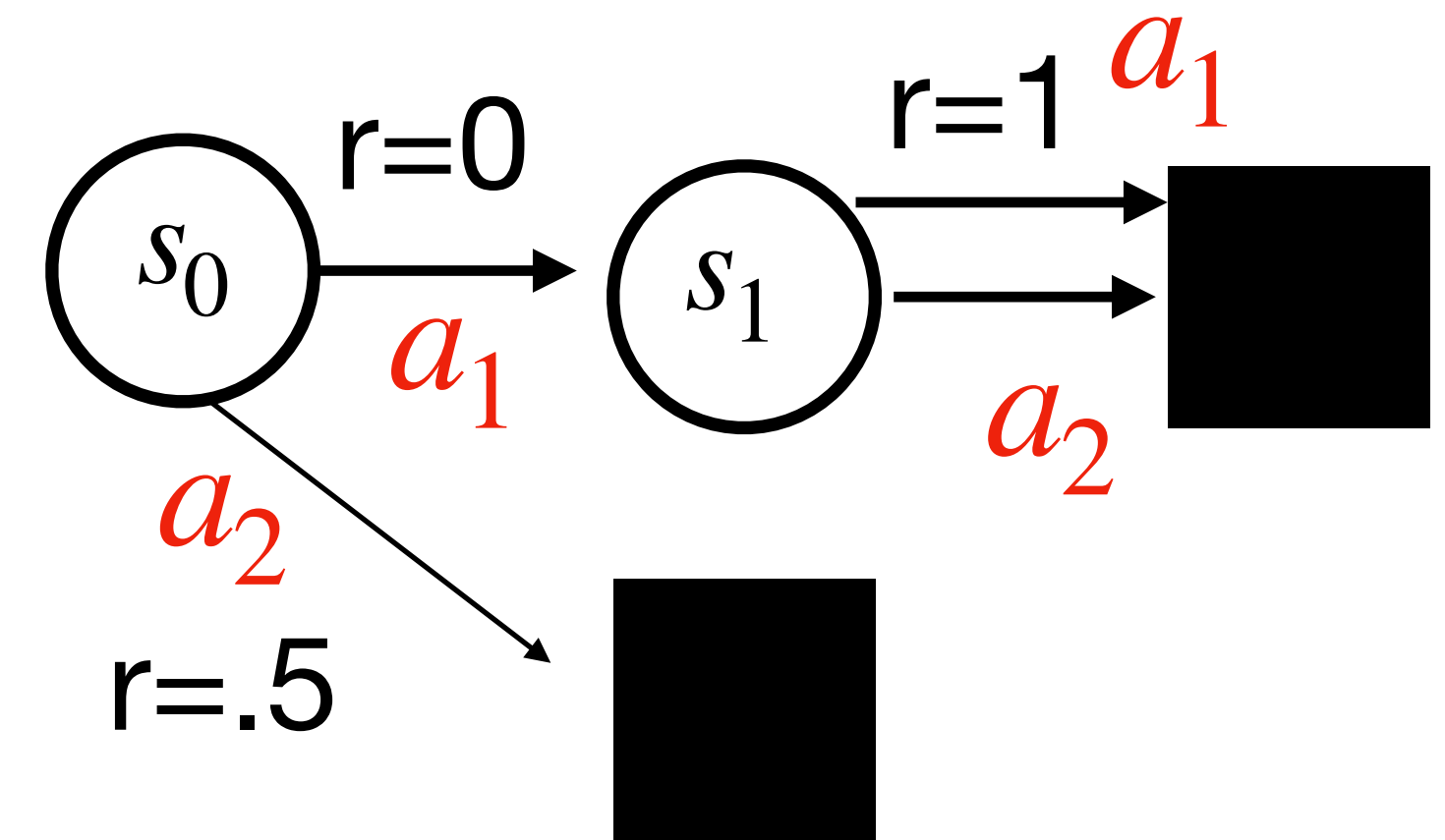
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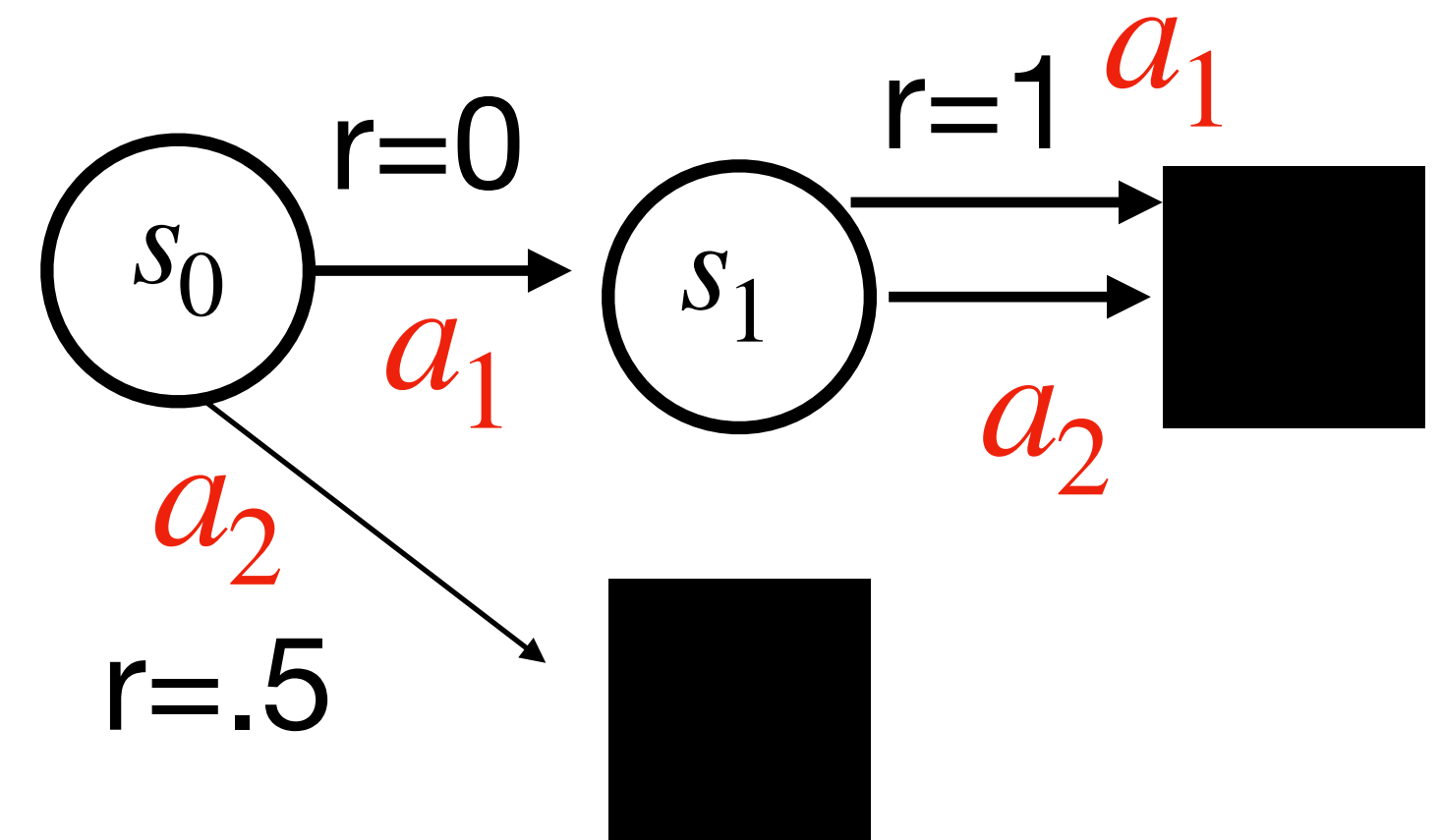
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Recap: Finite horizon Markov Decision Process

$$\mathcal{M} = \{S, A, r, P, H, \mu_0\},$$
$$r : S \times A \mapsto [0,1], H \in \mathbb{N}^+, P : S \times A \mapsto \Delta(S), s_0 \sim \mu_0$$

Note that in finite horizon setting, we will consider **time-dependent policies**, i.e.,

$$\pi := \{\pi_0, \pi_1, \dots, \pi_{H-1}\}, \pi_h : S \mapsto A, \forall h$$

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Policy interacts with the MDP as follows:

$$\tau = \{s_0, a_0, s_1, a_1, \dots, s_{H-1}, a_{H-1}\}, s_0 \sim \mu_0, a_0 = \pi_0(s_0), s_1 \sim P(\cdot | s_0, a_0), a_1 = \pi_1(s_1), \dots$$

Finishing Up From Last Lecture



1: An Iterative Algorithm: Policy Iteration



2: Convergence? How fast?

3: A new model: Finite horizon MDP

Compute Optimal Policy via DP

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The need for Exploration in RL:

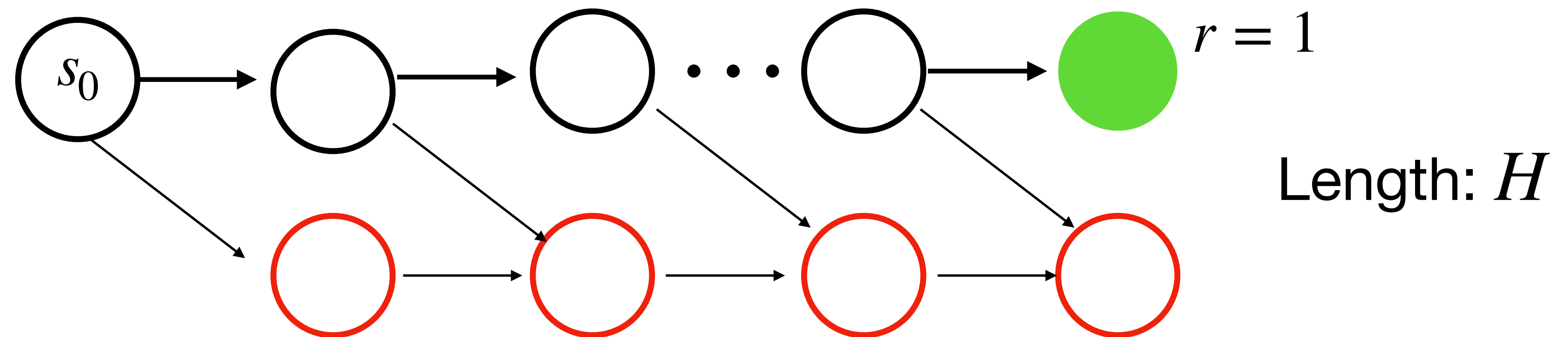
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The Combination Lock Example (i.e., the sparse reward problem)

- (1) We have reward zero everywhere except at the goal (the right end);
- (2) Every black node, one of the two actions will lead the agent to the dead state (red)



What is the probability of a random policy generating a trajectory that hits the goal?

Exploration!

We need to perform systematic exploration,
i.e., remember where we visited, and purposely
try to visit unexplored regions..

What we will do here:

Study Exploration in a very simple MDP:

$$\mathcal{M} = \{s_0, \{a_1, \dots, a_K\}, H = 1, R\}$$

i.e., MDP with one state, one-step transition, and K actions

This is also called Multi-armed Bandits

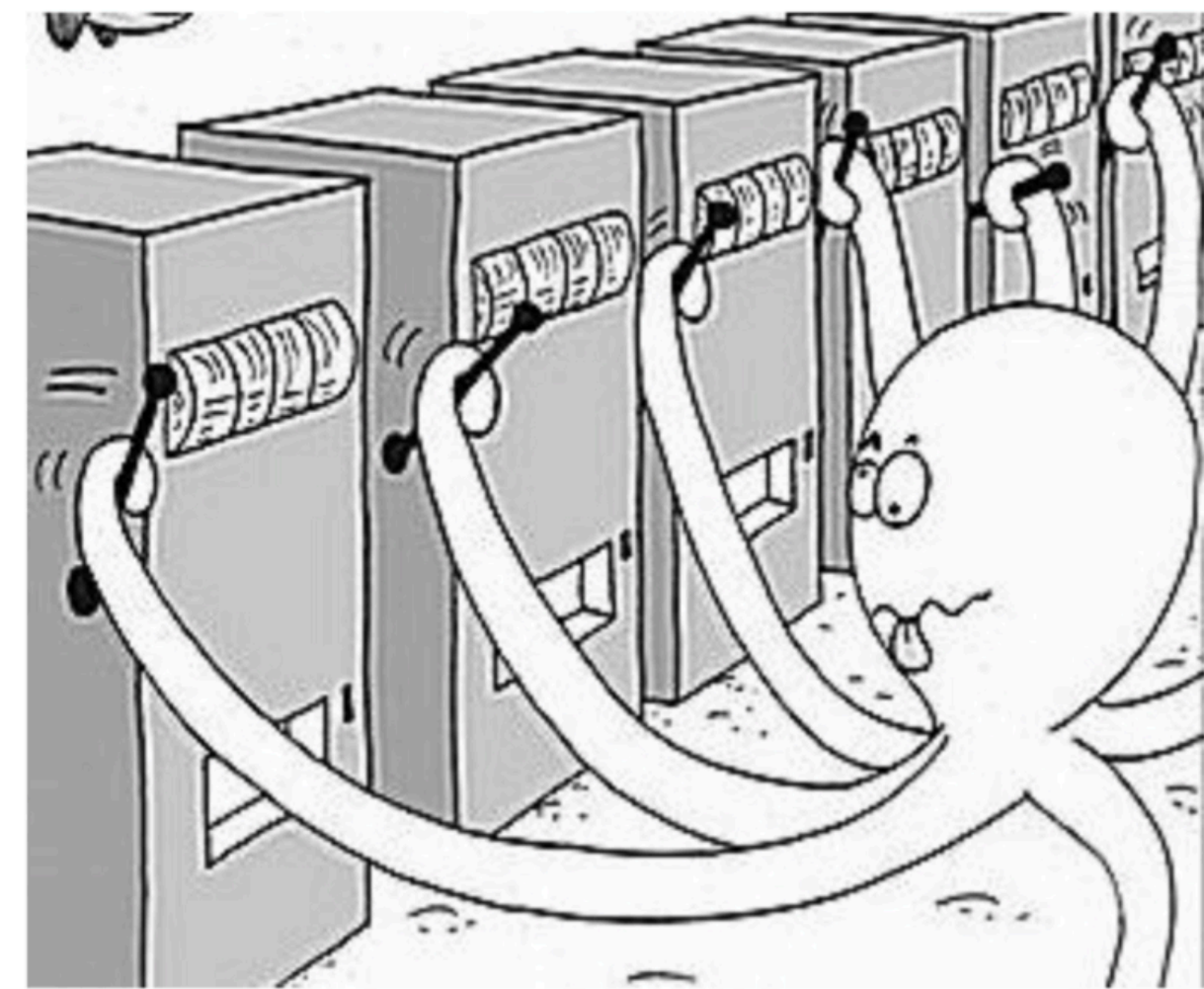
Outline

1. Introduction of MAB
2. Attempt 1: Greedy Algorithm (a bad algorithm)
3. Attempt 2: Explore-Then-Commit

Intro to MAB

Setting:

We have K many arms: $1, \dots, K$

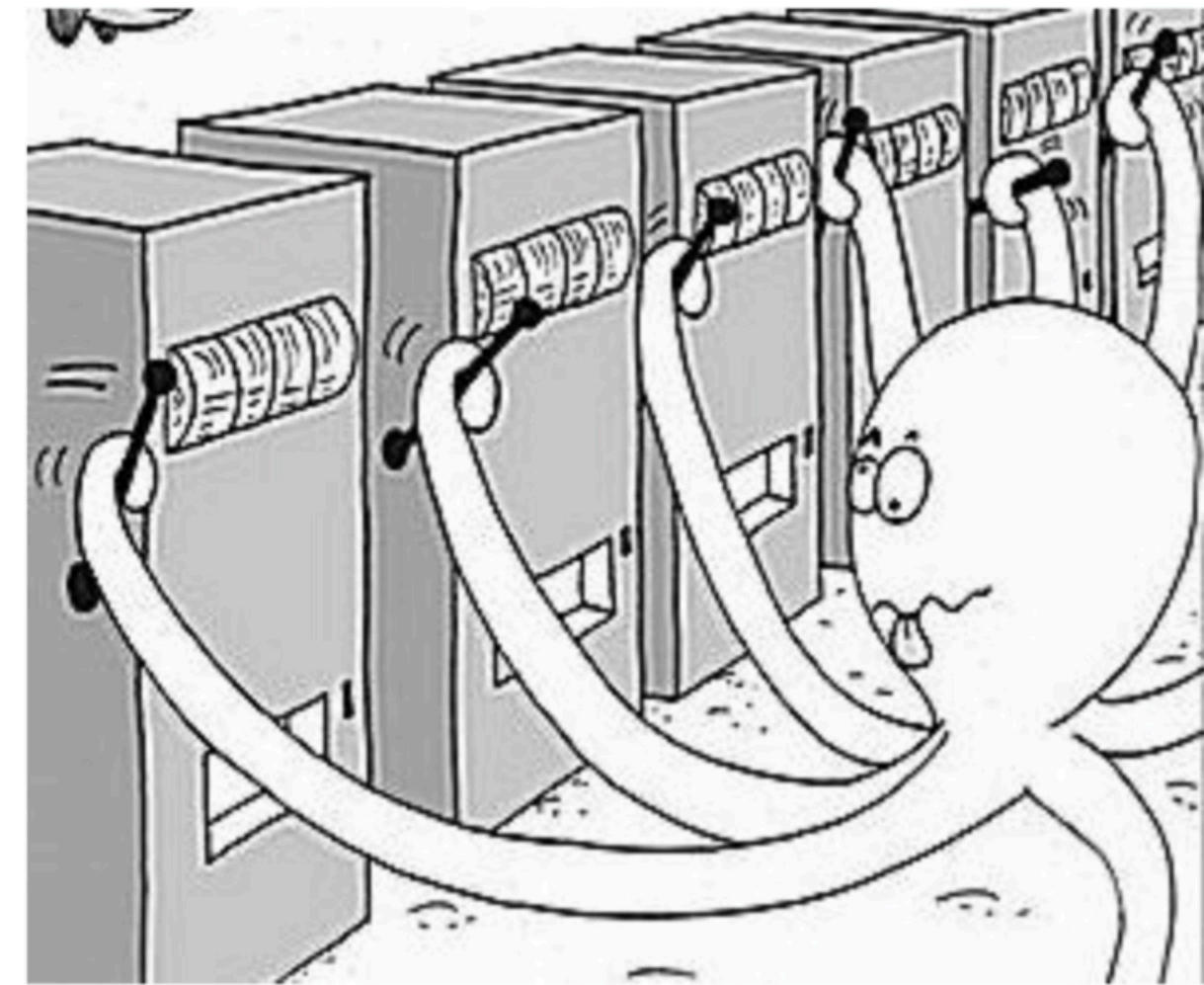


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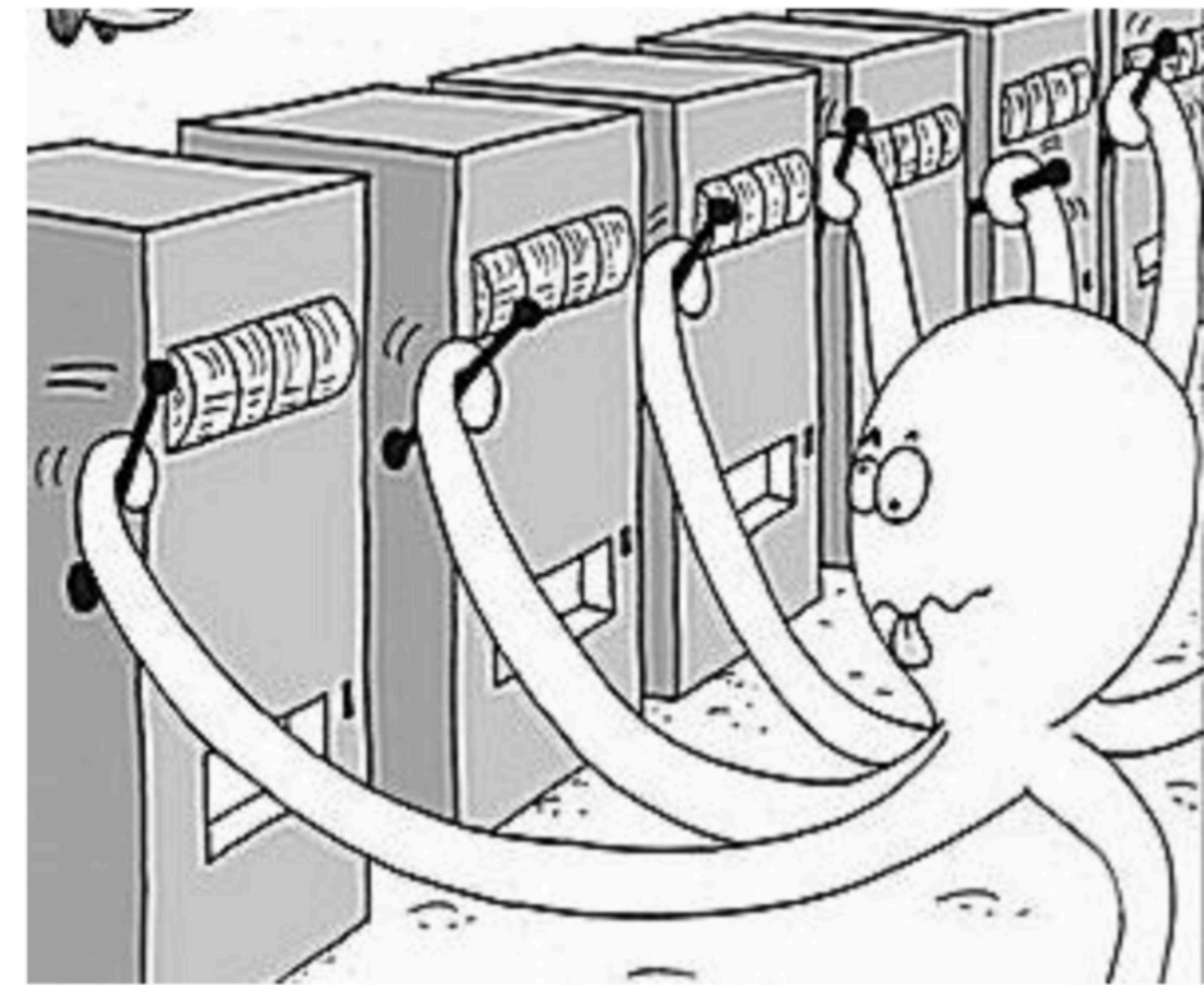
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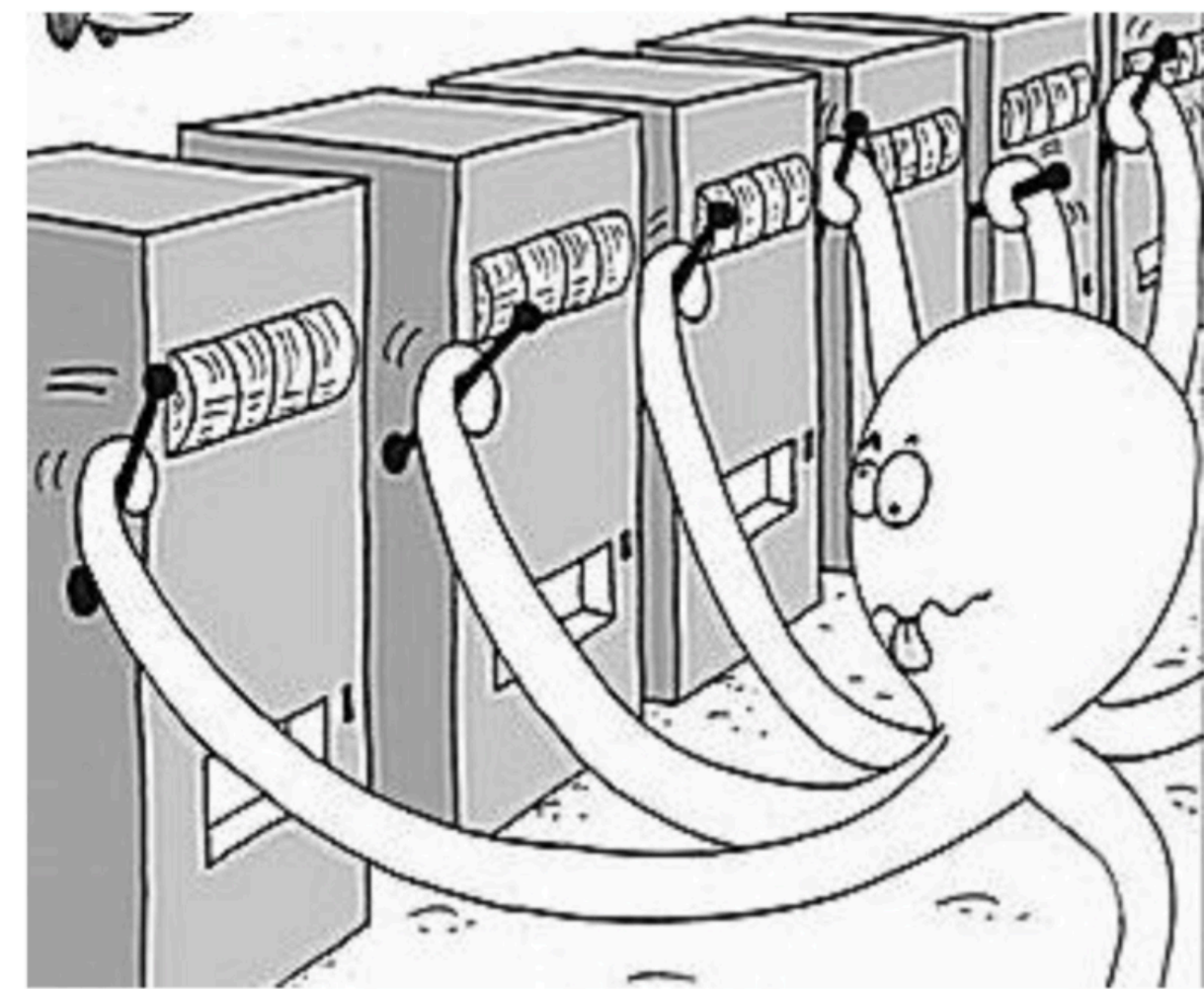
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Every time we pull arm k , we observe an i.i.d reward $r = \begin{cases} 1 & \text{w/ prob } \mu_k \\ 0 & \text{w/ prob } 1 - \mu_k \end{cases}$



Intro to MAB

Applications on online advertisement:



Arms correspond to Ads

Each arm has **click-through-rate**
(CTR): probability of getting clicked
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3. **Update**: Decide what ad to recommend for next round

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Why not sum the r_t ?

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Exploration and Exploitation Tradeoff:

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Exploration and Exploitation Tradeoff:

Every round, we need to ask ourselves:

Should we pull arms that are less frequently tried in the past (i.e., **explore**),
Or should we commit to the current best arm (i.e., **exploit**)?

Outline



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A bad arm (i.e., low μ_i) may generate a high reward by chance!
(recall we have $r \sim \nu$, i.i.d)

Attempt 1: Greedy Algorithm

More concretely, let's say we have two arms a_1, a_2 :

Reward dist for a_1 : w/ prob 60%, $r = 1$; else $r = 0$

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The greedy alg will pick a_2 —loosing expected reward 0.2 every time in the future

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(a bad algorithm: constant regret)

3. Attempt 2: Explore-Then-Commit

- Algorithm
- Analysis

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Yes, let's (1) try each arm multiple times, (2) compute the empirical mean of each arm, (3) commit to the one that has the highest empirical mean

How many times do we need to try each?

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Algorithm hyper parameter $N < T/K$ (we assume $T \gg K$)

For $k = 1 \rightarrow K$: (# Exploration phase)

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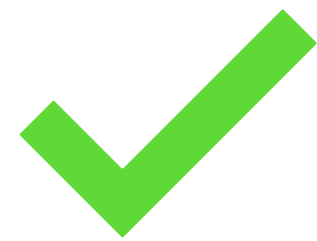
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Q: how to set N ?

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i.e., this gives us a confidence interval:

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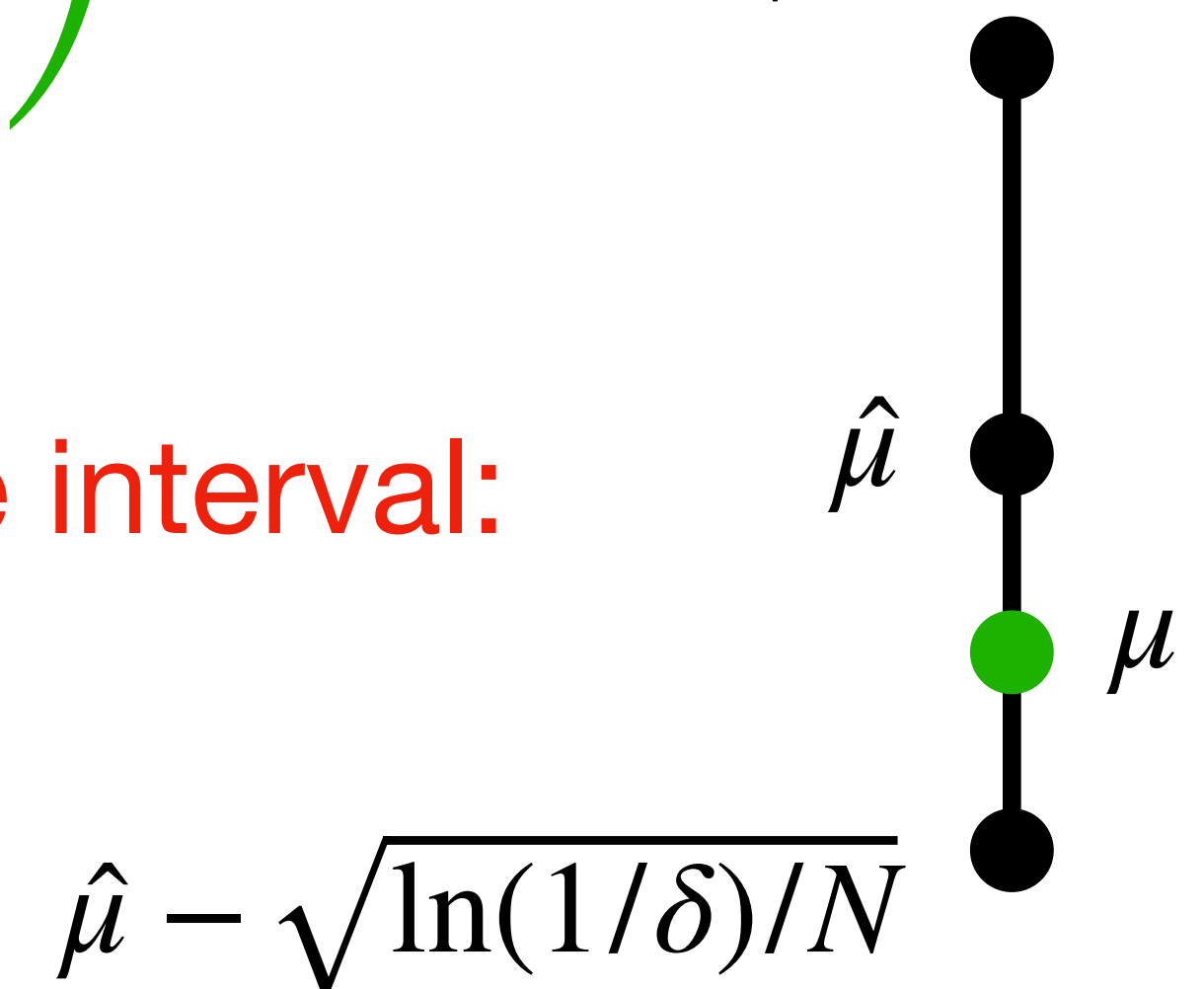
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$\{r_i\}_{i=1}^N \sim \mu$, w/ probability at least $1 - \delta$, we have:

$$\left| \sum_{i=1}^N r_i / N - \mu \right| \leq O \left(\sqrt{\frac{\ln(1/\delta)}{N}} \right)$$

$$\hat{\mu} + \sqrt{\ln(1/\delta)/N}$$

i.e., this gives us a confidence interval:



Statistical Tools:

2. Union Bound (optional):

$$\mathbb{P}(A \text{ or } B) \leq \mathbb{P}(A) + \mathbb{P}(B)$$

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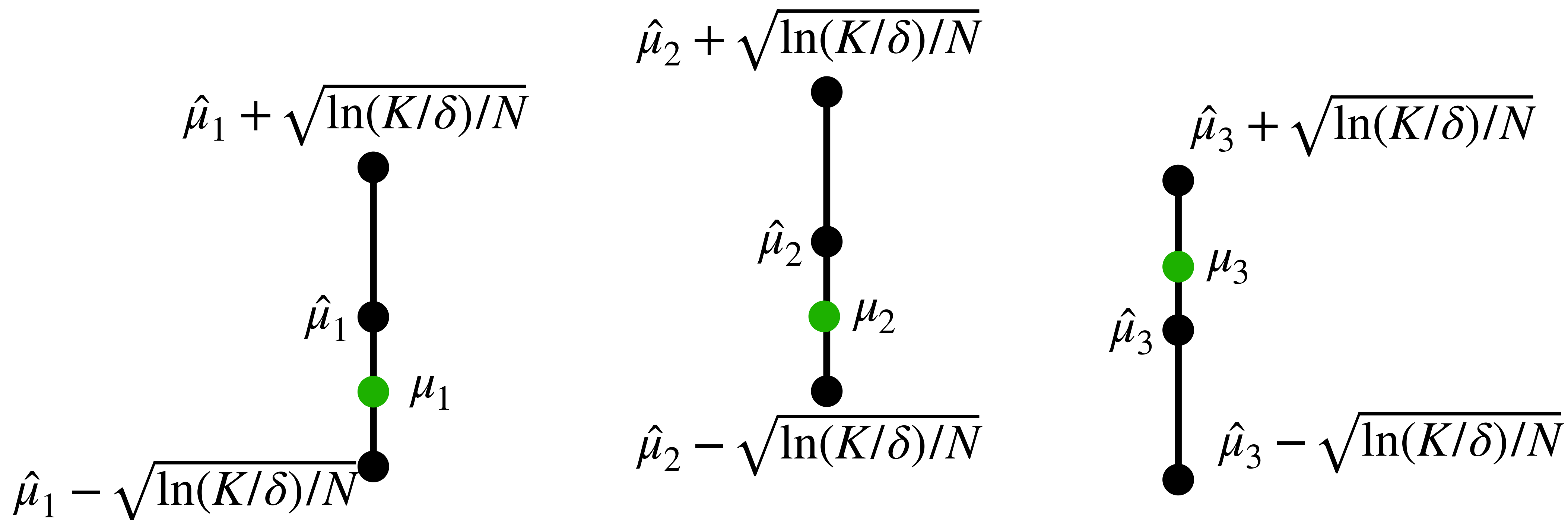
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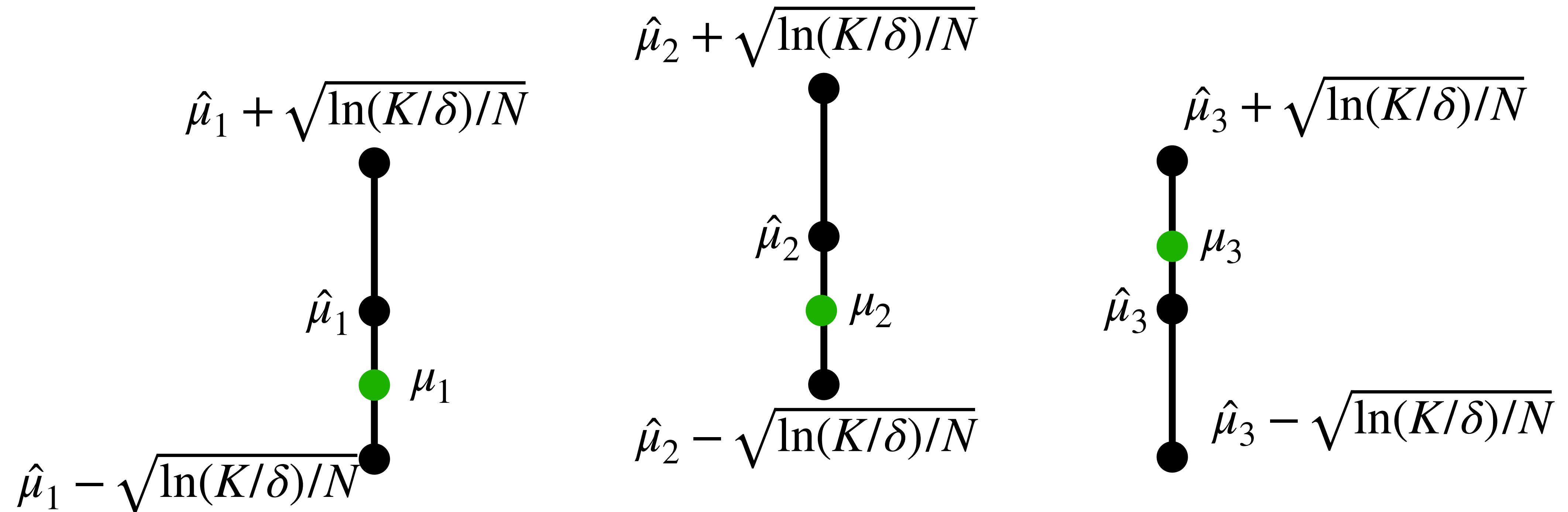
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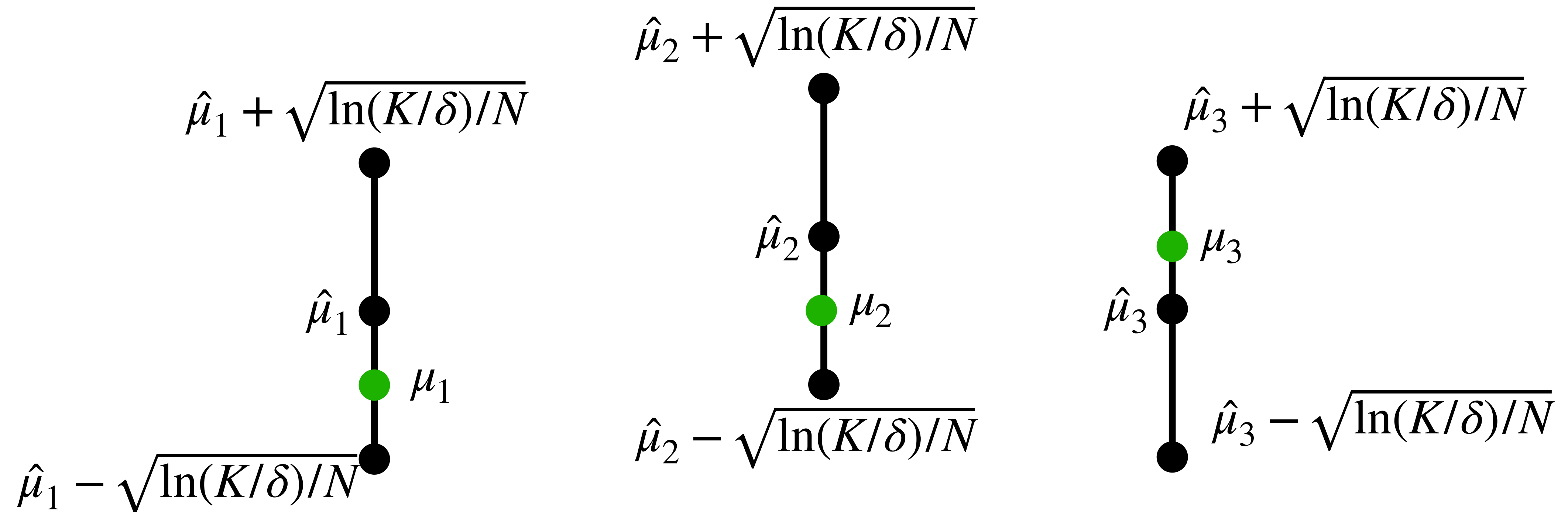
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Recall the Alg in this case will pick $a_t = 2$, for all $t \geq NK$,
 (but it will suffer regret $(T - NK)(\mu_3 - \mu_2)$)

Calculate the final regret:

Denote empirical best arm $\hat{k} = \arg \max_{k \in [K]} \hat{\mu}_k$, and THE best arm $k^\star = \arg \max_{k \in [K]} \mu_k$

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Let's now bound $\text{Regret}_{\text{exploit}}$

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$$\leq 2\sqrt{\ln(K/\delta)/N}$$

$$\text{Regret}_{\text{exploit}} \leq (T - NK)(\mu_{k^\star} - \mu_{\hat{k}}) \leq 2T\sqrt{\frac{\ln(K/\delta)}{N}}$$

Finally, combine two regret together:

$$\text{Regret}_{\text{explore}} \leq N(K - 1) \leq NK$$

$$\text{Regret}_{\text{exploit}} \leq (T - NK)(\mu_{k^*} - \mu_{\hat{k}}) \leq T \sqrt{\frac{\ln(K/\delta)}{N}}$$

$$\text{Regret}_T = \text{Regret}_{\text{explore}} + \text{Regret}_{\text{exploit}} \leq NK + 2T \sqrt{\frac{\ln(K/\delta)}{N}}$$

Minimize the upper bound via optimizing N:

$$\text{Set } N = \left(\frac{T \sqrt{\ln(K/\delta)}}{2K} \right)^{2/3}, \text{ we have:}$$

$$\text{Regret}_T \leq O \left(T^{2/3} K^{1/3} \cdot \ln^{1/3}(K/\delta) \right)$$

To conclude:

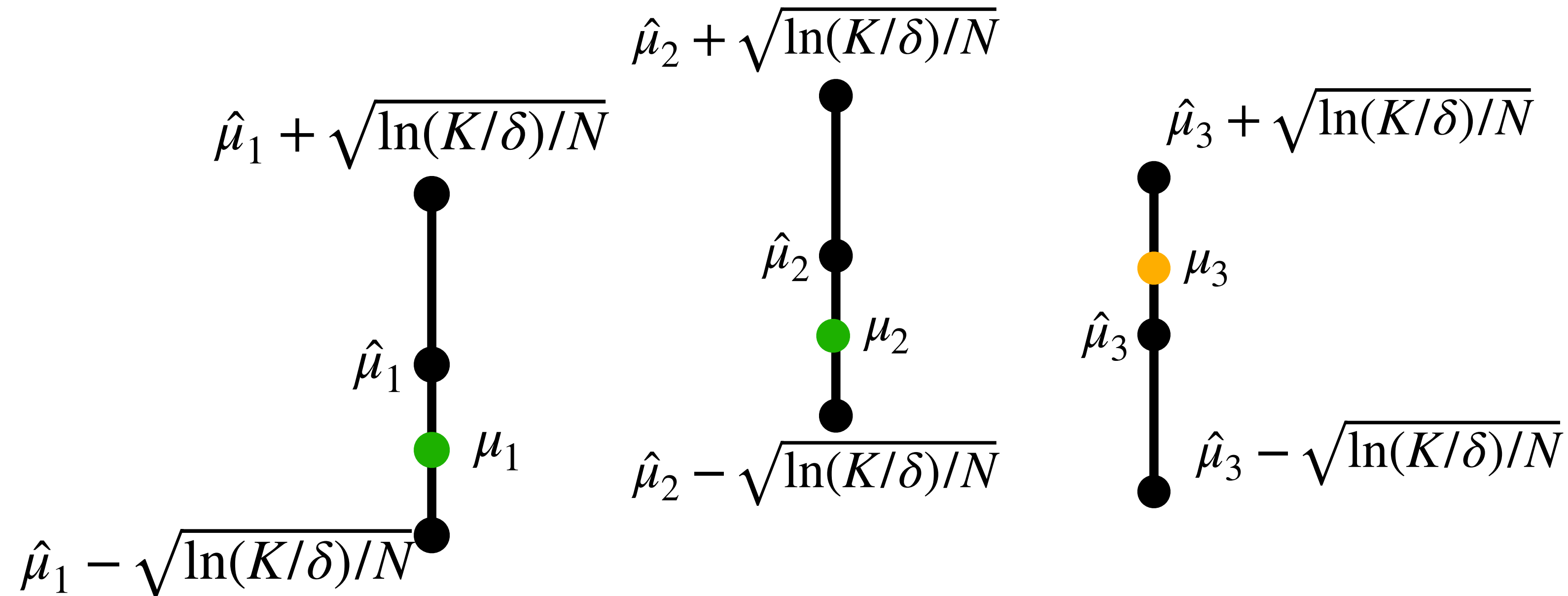
[Theorem] Fix $\delta \in (0,1)$, set $N = \left(\frac{T\sqrt{\ln(K/\delta)}}{2K} \right)^{2/3}$, with probability at least $1 - \delta$, **Explore-Then-Commit** has the following regret:

$$\text{Regret}_T \leq O \left(T^{2/3} K^{1/3} \cdot \ln^{1/3}(K/\delta) \right)$$

(See the MAB reading material for more details)

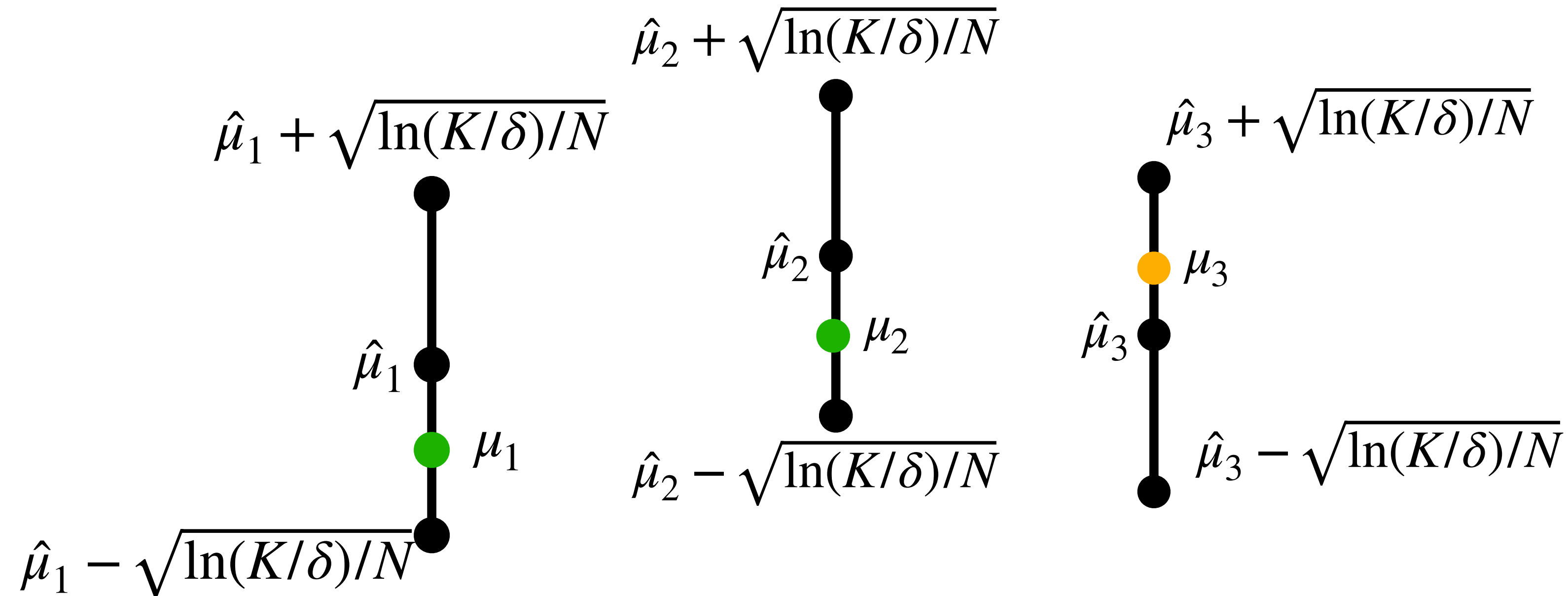
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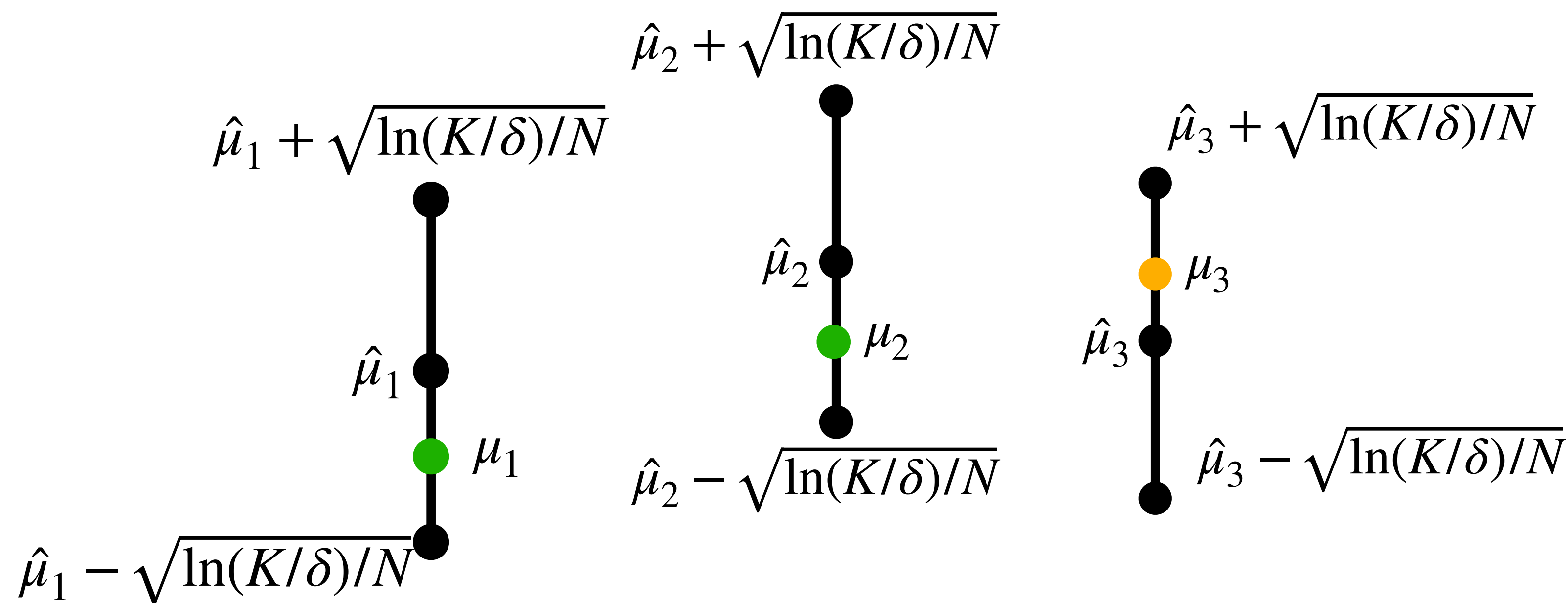
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But from the picture, we see that $(\mu_3 - \mu_2) \leq \text{length-of-Confidence-Interval}$

Summary

Attendance

<https://bit.ly/46pCUdN?r=qr>



Feedback

<https://bit.ly/4gYn2nC>

